

Seminar on Vagueness and Indeterminacy
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Introduction: The Sorites Paradox (CD)
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1 *Some cranky advice*

- This area of philosophy is especially plagued by ‘isms’. These are really loose clusterings of works held together by things like ‘a tendency to talk in such-and-such way’ or ‘a conviction that such-and-such concept is important’. Don’t waste your time trying to define the ‘isms’, and don’t expect them to line up with the important choice points that will emerge from a careful framing of the puzzles!
- The literature in this area also makes heavy use of certain jargon words—or rather, special quasi-technical usages of ordinary words. These include ‘vague’ and ‘indeterminate’, along with ‘borderline’, ‘definite’, ‘clear’, etc. Everything about these usages is contested, and there is great potential for pointless verbal disputes. The most fundamental questions in this domain are those that can be posed without using any of these difficult expressions—or at least, without *presupposing* an understanding of them.
- Be especially wary of the expression ‘there is a fact of the matter about whether...’. Some treat it as obvious that, e.g., there are people such that there is no fact of the matter about whether they are bald. And some treat ‘there is a fact of the matter about whether P ’ as obviously equivalent to ‘ P or not- P ’. But if you combine these approaches you can easily end up contradicting yourself: ‘there is no fact of the matter about whether x is bald’ should be equivalent to ‘it is not the case that (x is bald or x is not bald)’, which in turn seems equivalent to the outright contradiction ‘ x is not bald, and x is not not bald’.
- Although the topic of vagueness is standardly assigned to the philosophy of language, it’s not clear that this is right. The most famous puzzle in the area, the Sorites paradox, doesn’t on its face have anything to do with language. Of course, as with any philosophical puzzle, reflection on the language we use in setting it up might turn out to be helpful in solving it. But for some reason, the literature is extra-sloppy about explaining the relevance (if any) of theoretical claims about language to puzzles not about language.
- Be careful about weasel words like ‘claim’ and ‘statement’, which philosophers sometimes appeal to when they aren’t clear whether they are talking about linguistic items or propositions. Be careful too about the practice of displaying sentences with numbers or labels which are subsequently used as names—it’s easy to waffle about whether the labels work like names for sentences or like names for propositions.

- There'll be a lot of talk about “paradoxes”, such as the Sorites. Paradoxes are sometimes defined as arguments that seem valid, all of whose premises seem true, but whose conclusion seems false (or absurd). This suggests three natural paths to restore one's intellectual balance in the face of the threat posed by a paradox: one could refuse to accept at least one of the premises; one could accept the conclusion; or one could refuse to accept the validity of the argument. But this way of setting things up tends to stack the decks in favour of the last option—‘valid’ is after all a rather challenging term of art, which generations of logic instructors have struggled to explain to their students.

But for many so-called paradoxes (including the Sorites) the concept of validity is arguably a of a red herring. Suppose for example that we have a paradoxical argument that fits the following argument-scheme (the paradigm of validity for the Aristotelian tradition):

Barbara Every F thing is G .
 Every G thing is H .

 Every F thing is H .

Maybe we could live with the claim that this isn't valid. But it's quite another thing to steadfastly and clear-eyedly sustain a mental state in which, for certain F , G , and H , one is very confident that every F thing is G and that every G thing is H without being even moderately confident that every F thing is H . Insofar as one can imagine a person in this mental state at all, one seems to be imagining some kind of pathology, or a temporary and unstable position that might arise for someone who hasn't yet noticed the puzzle. No amount of discussion about *validity* per se will get us to the point where such a state would seem like a stable intellectual resting point, of the kind we are hoping to find when gripped by a paradox.

2 Tolerance and boundedness

Here are some handy definitions, where F is any property of natural numbers:

- F is **tolerant** iff every F number has an F successor.¹
- F is **bounded** iff some F number does not have an F successor.
- F is **intolerant** iff F is not tolerant
- F is **unbounded** iff F is not bounded.

¹ The **successor** of a natural number is the result of adding 1 to it.

The Sorites Paradox is often equated with arguments like the following, where for concreteness, let ‘N’ abbreviate ‘100’ let ‘small’ abbreviate ‘much smaller than 100’.²

1. Smallness is tolerant (i.e. every small number has a small successor).
 (Premise)

² Many other definitions could be substituted for these—e.g., you could take ‘N’ to abbreviate ‘100,000’ and ‘small’ abbreviate ‘such that necessarily any person with that number of hairs is bald’.

2. 0 is small. (Premise)
3. N is small. (From 1 and 2)

It's very hard to sustain the view that this isn't valid, and harder still to sustain a position in which one believes both premises (with high confidence) without believing (or even having moderate confidence in) the conclusion.

Note: some authors define ' F is tolerant' instead as 'Every number is such that if it is F , it has an F successor'. Or they focus on an argument whose major premise is a conjunction of conditionals 'If 0 is small, 1 is small, and...if $N-1$ is small, N is small'. These versions introduce what seem to me to be distracting considerations relating to the theory of conditionals. E.g., Priest (2010) thinks that there are many readings of 'if', some of which make the argument valid and others of which make it invalid, and the relevant premise is plausible only on certain readings in the latter category.³ But it is much less plausible that there is any relevant ambiguity in 'Every F number is G '.⁴

Our Sorites argument gives us a coarse-grained classification of theoretical options:

- *Wild* theorists (Unger 1979) believe 3 or reject 2.⁵ I'll henceforth ignore such views, which as far I can see lead to complete intellectual collapse.
- *Soritical* theorists believe 1 as well as 2, but still reject (i.e. refrain from believing) 3.
- *Orthodox* theorists reject 1 as well as 3, but still believe 2.

Being a (reflectively stable) soritical theorist requires a *massive* dose of logical unorthodoxy.⁶

3 Some choice points for the orthodox

Let's assume we are willing to be orthodox (like Kit and me), and moreover that we are happy to classify the above Sorites argument as valid. To illuminate the further choice points, consider the following argument, which is also classically valid.

1. 0 is small. (Premise)
2. N is not small. (Premise)
3. Smallness is intolerant (i.e. not every small number has a small successor) (from 1 and 2)
4. Smallness is bounded (i.e. some small number does not have a small successor) (from 3)

³ E.g., one that makes 'if P , Q ' equivalent to 'Either not- P or Q '. On this interpretation, the Modus Ponens inference ('If P , Q ; P ; therefore Q ') stands or falls with Disjunctive Syllogism ('Either P or Q ; not P ; therefore Q '), an argument-form which paraconsistentists like Priest regard as invalid.

⁴ Of course, 'every' is plausibly context-sensitive because of the well-known phenomenon of *quantifier domain restriction* ('Every beer is in the fridge'). But this seems entirely irrelevant here, since the intended domain clearly includes all the natural numbers.

⁵ I use the numerical label '2' to abbreviate 'that 0 is small', and similarly for '1', '3'. I'm never referring to the sentences.

⁶ Set aside views like that of Smith 2008 where one says that the argument is valid despite having high credences in both premises and low credence in the conclusion.

This gives a three-way classification of orthodox theorists. *Weakly orthodox* theorists reject 3; *moderately orthodox* theorists (like Kit) accept 3 but reject 4; *strongly orthodox* theorists (like me) accept both 3 and 4.

Both the weakly and moderately orthodox theorists can motivate their approach by appeal to the intuitive repugnance of the claim that smallness is bounded. This intuitive repugnance seems firmer than any intuitive attraction of the claim that smallness is tolerant.⁷

Both the weakly and moderately orthodox options involve significant logical revisionism.

- The moderately orthodox theorist must deny the validity of this very natural inference rule:

$$\text{Not Every/Some} \quad \frac{\text{Not every } F \text{ is a } G}{\text{Some } F \text{ is not a } G}$$

I find this hard to stomach. Very intelligent people seem to reason like this all the time: it's hard to believe they are just falling victim to a fallacy analogous to *affirming the consequent*!

- The weakly orthodox theorist must deny certain *metarules* that hold for validity in classical logic, e.g.:

Contraposition If the argument from $A_1...A_n$ and B to C is valid, the argument from $A_1...A_n$ and not- C to not- B is valid.

Reductio If the argument from $A_1...A_n$ and B to C and not- C is valid, the argument from $A_1...A_n$ to not- B is valid.

Evaluating these metarules gets us thoroughly into the difficult terrain of ‘what is this thing called “validity”, anyway’, so giving them up might seem a rather mild cost. To better assess the weakly orthodox position, I find it useful to look at a direct step-by-step argument for the intolerance claim, using very natural rules (comparable to 3) for the binary quantifiers ‘every’, ‘no’, and ‘some’:

1. Every tolerant property of zero applies to 100 (premise).
2. No tolerant property of zero doesn't apply to 100 (from 1).
3. No property of zero that doesn't apply to 100 is tolerant (from 2).
4. Every property of zero that doesn't apply to 100 is intolerant (from 3).
5. Being small is a property of zero that doesn't apply to 100 (premise).
6. Being small is intolerant (from 4 and 5).

Again, the weakly orthodox theorist has a choice of a few different places to get off the boat. All look...hard to get used to, quite apart from any general theorizing about the nature of validity.

⁷ The weakly or moderately orthodox theorist will be on even firmer ground if they also say that smallness is *not* bounded (as Kit does). This does however require a more radical logical revision than versions of the views which neither accept that smallness is bounded nor accept that it isn't.

4 Classical logic and theoretical conservatism

5 The challenge of dissolution

All approaches to the paradox face what Kit calls the challenge of *dissolving* the paradox by “explaining why we are taken in”.

- Views that accept that smallness is bounded need to explain why it strikes so many people that it would be absurd, unreasonable, weird to believe that this is the case.
- Views that reject some classically valid inferential step need to explain why it strikes so many people that it would be absurd, unreasonable, weird to believe the premises of the inference while reflectively withholding one’s assent from the conclusion.

The hope is often for the sort of explanation that could double as a kind of *therapy*. Understanding the source of one’s discomfort with a certain combination of attitudes can, sometimes, alleviate the discomfort.

Many attempts at dissolution don’t seem very satisfactory even as psychological explanations, never mind as therapy.

- Simple intellectual history—e.g. ‘Aristotle just made a blunder here and the whole philosophical tradition went along with him because he seemed so obviously a genius’—doesn’t really cut it. Wasn’t it *predictable* that we’d find ourselves with this puzzle?
- Non-classical theorists sometimes approach the dissolution task by pointing to some other inference rule, similar to the one they reject but with an extra premise, and saying that that rule really is valid, explaining our tendency to reason according to the original rule by appeal to our tendency to take the extra premise for granted. One candidate for the extra premise is (some relevant instance of) excluded middle (P or not- P). But this just pushes us back to the task of explaining why we are so tempted by this extra premise. This approach is particularly unpromising if the extra premise is an excluded middle instance, since even those who accept excluded middle generally admit that the relevant instances don’t have that much *pre-theoretic* appeal.
- Classical theorists sometimes introduce some theoretical terminology for talking about vagueness, such as a ‘determinately’ operator, and attribute our discomfort with the boundedness claim to our realization that no *determinately* small number has a successor that is *determinately* not small, combined with a posited mistaken tendency to infer from this supposed truth to the falsehood that no small number has a successor that is not small. But it seems mysterious why one would have such a tendency, or at least why the tendency would remain when one is fully cognizant that the relevant words are vague.

6 An exhaustive map?

References

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